REP-101 · Review under rules 17: balance, exact rules, screens and text, with a proposal

Game	Fruit Market 1.19.1 live (rules 16); rules 17 on branch `ccr-d8bcd65d-gblqqb`, not released
Report date	2026-11-11 (Asia/Kuala_Lumpur)
Type	Planning report (research and proposal). It approves no game change
Request	"with the latest rules, simulate and also playtest the live game ... to find balance changes to make the game better ... there is no mechanism if players cash reach 0 ... shopkeepers ... fruits ... upgrades ... favours ... events ... while maintaining or improving the game's strategic aspect, the game's bluff and social interaction, reducing memorising ... ui/ux ... cards need to be as practically close as it can be to the physical copy ... card portraits needs to be centred ... all text on card and in app needs to be audited as it feels too generated by AI ... prompt me for any required decision, then regenerate the print kit afterwards"; then "also analyse all 36 customer quantities/prices and all 20 delivery rows; contract shortages, rival allocation and fines when the winner cannot pay; Juice Bar interactions with contracts, Pasar Malam and final-day settlement; Premium Display versus Price Controls; exact tie resolution involving several special effects; amongst the rest"
Evidence	Four studies, kept whole in `docs/studio/review-17/`: `balance/BALANCE-FINDINGS.md` (about 60,000 computer games), `rules/EDGE-FINDINGS.md` (every case run in the engine), `screens/UX-FINDINGS.md` (35 findings, contact sheets in `screens/sheets/`), `text/TEXT-AUDIT.md` (1,636 strings)

1. How it was done

- **Simulation:** computer stalls at 2, 3, 4 and 6 stalls, Normal and Hard, 10 days (8 and 14 for upgrades). Limits: the computer does not buy for tomorrow's crowd, hold apples, bluff, or use Gossip and Scout well, so those are under-valued.
- **Correction:** the standard balance run (`npm run balance`) plays Hard computers in every seat. The figures in REP-096 and REP-100 are therefore Hard figures.
- **Playtest:** a full 10-day game on the live site (Day 1 by hand, the rest by a scripted player with every screen logged), a full rules 17 game, a 6-stall game, a run driven to $0, part games on a small phone and a desktop. No Quick match, Daily market or account on live.
- **Exact rules:** each interaction built as a position and run through the engine, then compared with RULES.md and the guidebook.
- **Text:** every player-facing string listed with a verdict.

2. What is wrong today, most important first

Live defects (1.19.1)

1. First game: while the "Open your stall" and "Secret prices" tips show, the whole table is squeezed into a strip at the left edge. Every new player meets it.
2. Tapping Daily market or Market League on the home screen replaces a game in progress with no question.

3. At 6 stalls the sales summary prints "undefined" for 5th and 6th place, and their result medals are blank.
4. How to play still says an unserved customer comes back the next day (removed in rules 14).
5. Mango Windfall delivers its "extra ripe mango" green, so the event only cuts mango prices.
6. Two Priority cards on one evening: the earlier one is wasted.
7. Big customer ticket cuts every face off at the eyes; small tickets cut the chin and the F O S lamps cover the right of the picture. (Left to right the faces are centred.)

Rules and balance

8. **Out of cash, out of the game.** A stall with under $3 and no fruit appears in 2% / 4% / 7% / 13% of games at 2 / 3 / 4 / 6 stalls and then finishes last 91 to 100% of the time. Causes: fruit that did not pay back 64% (typically green mangoes on day 1), standing-order fines 31%.
9. **A broke winner pays no fine.** The fine is taken before the rush and capped at cash. A stall can bid $1, spend its cash, deliver nothing, pay $0 and force rivals to sell at $1.
10. **Shopkeepers** (win rate against fair, Normal): Tip Jar −5 to −7 at every size; Nest Egg +6 to +9; Shade Cloth +6.6 and Bargain Hunter +4.7 at 3 stalls. The Haggler is weak only against Normal computers (a computer fault, not the card).
11. **Upgrades** (cash against never building, 10 days): Juice Bar +6.1, *PremiumDisplay* +4.7, Handcart 0, *RipeningRoom* −3.8, Order Book −$4.1.
12. **Fruit.** Mango is the default buy, not a gamble: $4.10 profit a piece against $2.40 to $2.90, 34 to 40% of all profit. Apple is the most wanted fruit and the only one delivered below demand; blueberry is the most over-supplied. No mango can be sold on days 1 and 2, yet 15 of 36 customers want one.
13. **Favours.** A stall earns about one a game (0.4 at 6 stalls). The deck never runs out. To a computer only Fridge (+4.55) *and* *FreeFruit* (+3.18) are worth money.
14. **Events.** None decides a game. Tourist Bus does nothing at 5 and 6 stalls; guests cost a stall money at 6 stalls (they replace a bigger walk-in); Blueberry Glut, Health Inspector and Price Controls barely matter.
15. **Memory.** A stall that does not remember what rivals bought wins 22.8% instead of 31.8%. That is bigger than any perk, and the computer remembers everything.
16. **Buying order** is the game's catch-up rule and it works: always first wins 58%, always last 10%. But Day 1 can hand the last buyer nothing sellable.
17. **Rule gaps:** Juice Bar pays on standing-order pieces but not on hawker or jam-maker pieces; Premium Display raises a guest's "fixed" price; Sharp Ears tips more than once a day and peeks when buying ends, not "each evening"; Nest Egg beats Priority on day 3; a Heatwave can take a green mango straight to Okay; the final tie falls to seat number; a rival's Okay piece can be taken before another rival's Fresh one; Favoured Trade is committed with prices in the app and after the flip in the guidebook.
18. **Checked and fine:** Price Controls beats Premium Display; the tie order in every combination; juice at night; the Closing Day order (rush, hawker, jam maker); order dates for each game length; a low bid is not a winning play for a stall with cash.

Screens and flow

19. Things a player must remember that the screen never shows: what rivals bought, today's buying order, how old their own pieces are, what they paid, what perks and upgrades do.

20. Steps with nothing to decide: "Done buying · 0 more allowed", Stocking with no ripe fruit, evenings with only "Lights out". About two thirds of a day is watching.
21. On the order's due day the order looks like a guest customer; a rival's piece is marked "sold out first"; the holder's summary never mentions the order or the fine.
22. Small phones (360 px): price rows, the bid controls and the rush result fall off the screen.
23. Results, Settings, Trophies, Wardrobe, How to play, the perk pick and the evening panel still have the old rounded look.
24. Every card type differs from the printed card, and the app and the print kit each carry their own wording.

Text

25. 1,636 strings: keep about 1,060, rewrite about 400, cut about 130. Exclamation marks 111 to 3. Three achievements can no longer be earned; several counts are out of date.

3. Proposal

Five packages, in this order. Each needs the owner's approval; publishing needs its own.

Package	What	Needs from the owner
A. Fix now (1.20.0)	Rules 17 as built, plus defects 1 to 7 above and a plain "You cannot afford anything" line. No balance change	Approval to publish
B. Rules 18: balance and rule gaps	The $0 net; fines after the rush; the tested number set; deck slips; rule gaps closed; Day 1; computer players fixed (Normal pricing, upgrade choice, buying for tomorrow)	Decisions D2 to D7
C. Text	One voice across cards and app; cuts; one table of card text shared by app and print	Decision D8
D. Screens and one card design	The same card in the app and on paper; memory aids; empty steps removed; small phones; old screens restyled	Mockups first, then approval
E. Print kit	The whole kit generated from the repo with the same cards as the app	After B, C and D

The tested number set (package B)

Change	Now	Proposed	Measured
Tip Jar	$2 when you serve a customer alone	$1 when a customer you sold to leaves with their whole order, $2 if you served them alone	−5.8 to −1.7 at 3 stalls
Nest Egg	Start with $18	Start with $15	+6.5 to +3.1 at 3 stalls
Upgrade costs	Juice Bar 4, Premium Display 6, Ripening Room 6, Order Book 5, Handcart 6	6, 5, 4, 3, 6	Best-to-worst gap $10.2 to $6.3
Ripening Room	Ripens in one night	Also: green mangoes cost $1 less	−2.05*to*+0.80
Order Book	Wins ties, halves fines	Also: an order pays you $2 more a piece	−2.22*to*+1.04

Change	Now	Proposed	Measured
Green mango	$3	$4	Profit a mango $4.10 to $3.30; share of profit 36% to 31%; blowouts 48.5% to 44.6%
Mango Windfall	Extra mangoes green	Ripe, as the card says	−2.33*to*−0.65 a stall
Normal computer	Prices as if it held the Haggler	Does not	Haggler −10.4 to −3.0

Whole set, largest distance from fair: 8.6 to 5.8 (2 stalls), 10.4 to 4.4 (3), 8.6 to 4.1 (4), 6.9 to 5.3 (6). Closer to the target of 4, not proof of it (each figure is ±2). Still out: Shade Cloth about +4 to +7 at 3 stalls (no number to trim), Sharp Ears −4.4 at 3 stalls.

Deck slips (package B)

- Coral Mei, Magic Kumar, Professor Pebble: blueberry $3 to $4 (a Soft one cannot make money at $3).
- Short Load, 3-stall row 2/1/1/1 to 2/2/1/1 (it equals the 2-stall row). Summer Run apples 1/1/1/1/1 to 1/1/1/2/2.
- Optional: one Premium watermelon customer (four cards are identical "3 watermelon at $7"); +1 apple on five lorries if apple should not be scarce.

4. Decisions

#	Decision	My recommendation
D1	Publish package A now, or wait for everything	Publish now: two of the defects hit every new player and can lose a saved game
D2	The $0 net	A stall with an empty shelf and less than $5 at dawn is topped up to $5. It cannot be reached by buying fruit. (Simpler but leanable: any stall under $3 topped up to $3)
D3	The number set, with or without the $4 green mango	With it: it is the one change that makes the mango a gamble
D4	Standing orders: fines and low bids	Fine paid after the rush; what cannot be paid is forgiven; lowest bid allowed is half the card's price, so rivals are never forced to sell for $1
D5	Should what rivals bought today stay on show until the bell	Yes: today's buys only, as pips on each rival's box (and face up beside the stall on a table). The bluff moves to what they put out, not to who remembers best
D6	Day 1	A fixed first lorry with ripe fruit for every stall, matched to the first crowd; no mango wanted before day 3
D7	Rule gaps (item 17)	Engine follows the card text in each case; listed in `rules/EDGE-FINDINGS.md` with the fix for each
D8	The voice of the text, and the setting	Plain market voice, one short line per character; keep "harbour market" unless the owner wants it Malaysian throughout
D9	Steps with nothing to decide	Buying ends by itself at the limit; Stocking is skipped with nothing ripe; an evening shows only if you have a choice

#	Decision	My recommendation
D10	Three achievements that cannot be earned	Replace them with ones that can

5. Not checked

- A real phone (touch, notch, sound). Online rooms on live. Pass-and-play hand-over.
- The printed PDFs as pictures (text only).
- Whether people lean on a $3 net (bounded at about $5 a game).
- The favour deck's make-up (effects under $1 a game with computers).
- `npm run verify` on the number set: on a clean copy 441 tests pass and 10 fail; eight pin old numbers, two need a look.